

Dita Pelaj

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🌐 linkedin.com/in/dita-pelaj | 🐙 github.com/ditaa99 | 🌐 Portfolio

Data Science master's student with a strong Computer Science background and research experience in Software Engineering and Front-End development. Passionate about data-driven problem-solving and continuously enhancing analytical skills. Additionally, interested in theoretical computer science, especially automata theory.

EDUCATION

University of Potsdam

Oct. 2024 – Present

MSc Data Science

Potsdam, Germany

- Built and trained ANN and CNN models to predict train delays from weather-related data
- Applied statistical analysis and built reproducible workflows using Python and Snakemake

University of Pécs

2021 – 2024

BSc Computer Science

Pécs, Hungary

- GPA: 4.57 / 5.0; Stipendium Hungaricum Scholarship awardee
- Thesis: *Educational Software for Formal Languages and Automata*

EXPERIENCE

University of Potsdam – Software Engineering Department

Oct. 2025 – Present

Research Assistant

Potsdam, Germany

- Project funded by the Center for Quality Development in Teaching and Learning (ZfQ) to enhance educational frameworks and course quality.
- Supervised student engineering projects, acting as an external client by defining requirements, change requests, and feedback for BSc students enrolled in the Software Engineering II course.

Tech Frame

Jul. 2023 – Aug. 2023

Front-End Intern

Prishtina, Kosova

- Developed user interfaces using HTML, CSS, JavaScript, and React.
- Integrated WordPress APIs with React applications to improve interactivity and responsiveness.
- Gained practical experience in debugging, API integration, and software development workflows.

Outdoor Kosova

Jul. 2017 – Aug. 2021

Tour Guide

Radavc, Peja, Kosova

- Guided groups of up to 30 visitors, developing communication and public-speaking skills.

SKILLS

Programming & Frameworks : Python, Java, C++, JavaScript, SQL, Flask, React, Spring Boot, Bootstrap

Databases: PostgreSQL, MySQL, Microsoft SQL Server

Data Science : TensorFlow, Scikit-learn, Pandas, NumPy, Statistical Analysis, Data Visualization, PCA, OLS

Tools : Git, Docker, Jupyter, Snakemake, Firebase, Fly.io, WordPress

Languages: Albanian (Native), English (Fluent), German (Intermediate)

PROJECTS

AgroClimate – Climate and Crop Analysis | *Python, Snakemake, Statistics*

- Built a reproducible pipeline combining climate (DWD) and crop yield (OpenAgrar) data for Brandenburg.
- Applied statistical models (OLS, PCA) and visualizations to assess climate impacts and forecast yields.
- Automated workflows, generating 100+ plots and summaries for reproducible research

Train Delay Prediction from Weather Data | *Python, Machine Learning*

- Developed ANN and CNN models to predict train delays from weather-related data
- Performed preprocessing, feature analysis, model training, and evaluation

GenLang – Formal Languages and Automata Tool | *React, Flask, Python*

- Built a web application that generates formal languages from regular grammars
- Implemented grammar validation, derivation generation, and language pattern display

Zurich Traffic Accident Analysis | *Python, Data Analysis, Visualization*

- Analyzed accident data to identify temporal, spatial, and categorical patterns
- Created visualizations exploring accident distributions by type, time, day, and location
- Applied statistical and correlation methods to identify factors influencing accident frequency